Inferential Models

Ryan Martin
North Carolina State University[1]
Researchers.One[2]

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University of Liverpool, UK
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1 www4.stat.ncsu.edu/~rgmarti3
2 https://researchers.one

Introduction

- Statistical inference is concerned with quantification of uncertainty about unknowns based on data, models, etc.
- The phrase "uncertainty quantification" is almost exclusively associated with *ordinary* probability.
- Therefore, statisticians associate uncertainty quantification with data-dependent probability distributions.
- However, now we know that ordinary probability
  - is not the only option,
  - and often not the best option.

Intro, cont.

- There are signs in the statistics literature that ordinary probability isn't right for inference:
  - Fisher's attempts to define *fiducial probability* for inference lost out to Neyman's notion of *confidence*.
  - Bayesian posteriors have good statistical properties only when the prior is "non-informative," a notion foreign to probability.
- Dempster's attempt to fix Fisher's fiducial argument was part of *imprecise probability*'s origin story.
- So, despite the strong desire to handle statistical inference entirely within the theory of probability, there's good reason to believe this isn't possible.

Intro, cont.

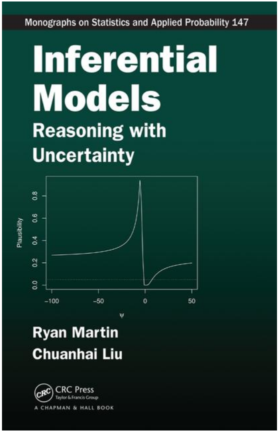
- What, then, is a more appropriate framework?
- Of course, it's not just a matter of switching from precise to imprecise probability, the formulation has to "work."
- Key questions:
    - In the context of statistical inference, what does it mean for uncertainty quantification to *work*?
    - How to construct something that *works* in the above sense?
- My goal here is to answer these questions.

This series of talks

- This is a three-part series of lectures.
- These lectures split (unequally) into three themes:
  - *Why and What* ($\approx$ 1 lecture)
  - *How* ($\approx$ 1.5 lectures)
  - *What's next* ($\approx$ 0.5 lectures)
- More specifically:
  1. Why do I define "works" the way I do and what are the consequences of this definition?
  2. Given the goal to construct something that "works" in this sense, how to actually do the construction?
  3. Open problems, things I'm currently thinking about, etc.

References

- Part 2 of the lecture is mostly from the book; you can also find the corresponding papers on my website.



References, cont.

- Parts 1 and 3 are mostly based on new material, to which I'll provide some references on the slides.
- You can find all of these references on my website.
- Even if the papers are published, I'll refer to versions on
  - `arxiv.org` or
  - `researchers.one` (`R1`)
- Why? Same content, but more easily accessible.
- Journals make accessibility (and other things) more difficult.
- Unfortunately, researchers need journals to be engaged in the scholarly community.[3]
- *Researchers.One* is trying to change that.

[3] How else to be sure that at least one person looks at our papers?

- Aims to integrate the various things that researchers do:
  - peer review and publications
  - conferences and proceedings
  - ...

RESEARCHERS.ONE

BFF 6.5 – Virtual Workshop on Bayesian, Fiducial, and Frequentist Statistical Inference

Date:	Friday, February 5th, 2021
Times:	Morning Session, 9:00–11:00am EST
	Afternoon Session, 1:00–3:00pm
Organizers:	Jan Hannig (University of North Carolina Chapel Hill)
	Ryan Martin (North Carolina State University)
Registration:	`https://researchers.one/conferences/bff65`

In memoriam

- Don Fraser passed away on Dec 21st, 2020, age 95.
- Recognized and wrote about the limitations of ordinary probability in the context of statistical inference.[4]
- His work made a tremendous impact on the field of statistics and on me personally.

[4] A favorite: Fraser (2011). Is Bayes just quick and dirty confidence? *Statistical Science*, with discussion and rejoinder.

Part 1: Outline

- Statistical problem setup
- Probabilistic inference
- Validity property and its consequences
- Questions:
  - Can probabilities be valid?
  - If not probabilities, then what?
- Frequentism, validity, and imprecise probability
- Summary and preview of Part 2

Statistical inference problem

- Observable data: Y in a sample space $\mathbb{Y}$.
- Data are assumed to be relevant to the phenomenon under investigation, formalized through a *model*.
- Statistical model: a collection of probability distributions

$$\mathscr{P} = \{\mathsf{P}_{Y|\vartheta} : \vartheta \in \Theta\}.$$

- I write ϑ for generic values of the parameter, and θ for the unknown true value.[5]
- Goal is to quantify uncertainty about the unknown θ based on observed value $Y = y$ of the data.

[5] Here I'll assume the model is correct, so there is a "true" parameter value. It's possible to relax this, see M. (2019), `R1/articles/18.08.00017`

Statistical inference problem, cont.

- Two mainstream schools of thought:
  - *frequentist*
  - *Bayesian*
- These two approaches are very different:
  - the former constructs procedures (hypothesis tests, confidence intervals, etc.) with "good properties"
  - the latter uses a prior distribution for θ and conditional probability theory to construct a posterior distribution.
- Having two accepted theories is problematic on many levels, it suggests important connections/insights are missing.
- My goal is to fill this gap.

Main references in Part 1

- M. (2019). "False confidence, non-additive beliefs, and valid statistical inference."[6]
- M. (2021)...[7]

RES3ARCHERS.ONE researchers.one/articles/21.01.00002

An imprecise-probabilistic characterization of frequentist statistical inference*

Ryan Martin†

January 25, 2021

[6] researchers.one/articles/19.02.00001

[7] researchers.one/articles/21.01.00002

Probabilistic inference

- By "uncertainty quantification" I mean a function that:
  - depends on data $Y = y$, assumed model, etc.
  - assigns to each assertion $A \subseteq \Theta$ a number in $[0, 1]$ that represents ones degree of belief in A, given data, etc.
- A *capacity* γ is a set function that satisfies:
  - $\gamma(\varnothing) = 0$ and $\gamma(\Theta) = 1$
  - If $A \subseteq A'$, then $\gamma(A) \leq \gamma(A')$.

Definition.

An *inferential model* (IM) is a mapping from $(y, \mathscr{P}, \ldots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^\Theta \to [0, 1]$ such that, for each $A \subseteq \Theta$, $\underline{\Pi}_y(A)$ represents ones degree of belief in A, given data, etc.

Probabilistic inference, cont.

Definition — copied from above.

An *inferential model* (IM) is a mapping from $(y, \mathscr{P}, \ldots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^\Theta \to [0,1]$ such that...

- Remarks:
  - The "..." allows for other inputs, e.g., prior information.
  - $\underline{\Pi}_y$ has a dual, $\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c)$, may not be distinct.
  - $\overline{\Pi}_y(A)$ represents the plausibility of A, given data, etc.
- Examples include:
  - Bayesian posterior distributions
  - fiducial and confidence distributions
  - generalized Bayes
  - Dempster's "(p, q, r)"
  - ...

Probabilistic inference, cont.

- The IM definition itself is too vague to be useful...
- To sharpen our focus, we need some additional constraints.
- Basic question: *what do we want to use an IM for?*
- We *don't* really need an IM for:
  - point estimation
  - isolated hypothesis testing
  - ...
- The motivation is "uncertainty quantification," getting an overall picture of how beliefs are assigned.
- That is, we care about the mapping $A \mapsto \underline{\Pi}_y(A)$.

Probabilistic inference, cont.

- One idea: require $\underline{\Pi}_y$ to be *coherent*.[8]
- But the statistics problem has a non-subjective element.
- Basic principle:

$$\text{“if } \underline{\Pi}_y(A) > t, \text{ then infer } A\text{”}$$

- Assertions about θ are *true* or *false*, and inferring a false assertion can have negative consequences.
- Idea: require $\underline{\Pi}_y$ to prevent these bad outcomes...?

[8] More on this idea in Part 3.

Probabilistic inference, cont.

- $\underline{\Pi}_y$ depends on data y that we have no control over, so it's not possible to entirely avoid the bad outcomes.
- However, we can prevent *systematically* bad outcomes.
- That is, of the mapping $y \mapsto \underline{\Pi}_y(\cdot)$, we can require that

$$\big(\theta \not\in A\,, Y \sim \mathsf{P}_{Y|\theta}\big) \implies \underline{\Pi}_Y(A) \text{ tends to be small.}$$

- This will be made more precise below.
- Consistent with the message in Reid & Cox:

  > *it is unacceptable if a procedure. . . of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions*

Valid probabilistic inference

Definition.

An IM is *valid* if, for all $A \subseteq \Theta$ and all $\alpha \in [0,1]$,

$$\sup_{\theta \notin A} \mathsf{P}_{Y|\theta}\{\underline{\Pi}_Y(A) > 1-\alpha\} \leq \alpha.$$

- Intuitively, validity controls the frequency at which the IM assigns relatively high beliefs to false assertions.
- Why the "$1-\alpha$" threshold on the inside?
  - wouldn't make sense if threshold depended on Y or on A
  - if the threshold can only be a (decreasing) function of α, then what else is there?
- Other questions about validity to be discussed in Part 3:
  - why for all $\alpha \in [0,1]$?
  - why for all $A \subseteq \Theta$?

Valid probabilistic inference, cont.

- There are (at least) two important consequences of validity.
- The first is that an equivalent definition can be given in terms of the capacity's dual, i.e., the IM is valid if

$$\sup_{\theta \in A} \mathsf{P}_{Y|\theta}\{\overline{\Pi}_Y(A) \leq \alpha\} \leq \alpha \quad \text{for all } (\alpha, A).$$

- The second is stated in the following theorem.

Theorem.

If the IM $(y, \mathscr{P}, \ldots) \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$ is valid, then:

1. "Reject $H_0 : \theta \in A$ if $\overline{\Pi}_y(A) \leq \alpha$" is a size α test.
2. The $100(1-\alpha)\%$ plausibility region

$$\{\vartheta : \overline{\Pi}_y(\{\vartheta\}) \geq \alpha\}$$

is a $100(1-\alpha)\%$ confidence region.

Valid probabilistic inference, cont.

- Validity provides protection against Cox & Reid's "systematically misleading conclusions"
- Validity is a strong condition, but I think there's good reason for it, which I'll explain at the end of Part 1.[9]
- The next step is to understand what kind mathematical structures in $A \mapsto \overline{\Pi}_y(A)$ are consistent with validity.
- In particular:
  - can additive probabilities achieve validity?
  - and, if not, then what can?

[9] I'll talk about possible relaxations in Part 3.

Satellite collision problem

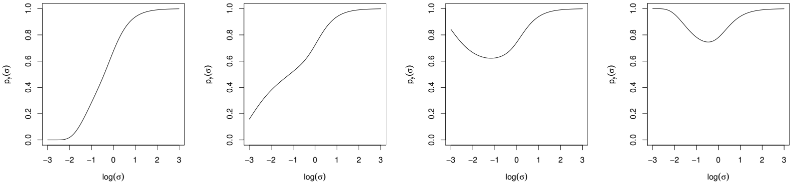
- A satellite orbiting Earth could collide with another object.
- Potential mess, so navigators try to avoid collision.
- Data on position, velocity, etc is used, along with physical and statistical models, to compute a *collision probability*.[10]
- If collision probability is low, then satellite is judged safe; otherwise, some action is taken.
- An unusual phenomenon has been observed in the satellite conjunction analysis literature:

noisier data makes non-collision probability large.

[10] Collision probability is based on a standard Bayes/fiducial/confidence distribution; see Balch, M., and Ferson (2019), `arXiv:1706.08565`.

Satellite collision, cont.

- Simple illustration:
  - $\|y\|$ denotes measured distance between satellite and object.
  - True distance ≤ 1 implies collision.
  - Measurement error variance is σ^2.
  - $p_y(\sigma) =$ probability of non-collision, given y.
- When σ is large, $p_y(\sigma)$ is large, no matter what y is!



(a) $\|y\|^2 = 0.5$ (b) $\|y\|^2 = 0.9$ (c) $\|y\|^2 = 1.1$ (d) $\|y\|^2 = 1.5$

False confidence

- Even if satellite is on a direct collision course, if there's enough noise, then probability tells navigators that it's safe.
- *False confidence:* a hypothesis tending to be assigned large probability even though data does not support it.
- Is this a general phenomenon...?

False confidence theorem (Balch, M., and Ferson 2019).

Let Π_Y be any additive belief on Θ, depending on data Y. Then, for any α and any t, there exists $A \subset \Theta$ such that

$$A \not\ni \theta \quad \text{and} \quad \mathsf{P}_{Y|\theta}\{\Pi_Y(A) > t\} > \alpha.$$

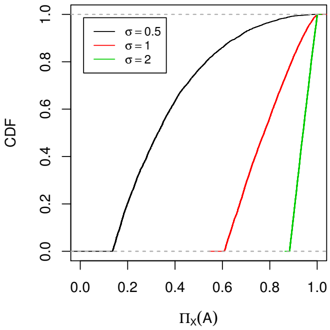
(This is assuming $\Pi_Y \ll$ Lebesgue, a similar result probably holds more generally.)

False confidence, cont.

- In words, theorem says there always exists false hypotheses that tend to be assigned high Π_Y-probability.
- *Therefore, an IM whose output is probability can't be valid!*
- More on this at the end of Part 1.

False confidence, cont.

- Simplified version of the satellite example.
  - Let $A = \{\text{non-collision}\}$.
  - Then $\Pi_Y(A)$ as a random variable, with a CDF
  - Plot CDF when data are generated under a collision course.
- *False confidence:* $\Pi_Y(A)$ is almost always large!



Frequentist to IM

- All of the "frequentist" statistical procedures you know are based on the following fundamental idea:
  - Let $T_{y,\vartheta}$ be some real-valued function of $(y, \vartheta) \in \mathbb{Y} \times \Theta$.
  - Assume $T_{y,\vartheta}$ is large iff y and distribution $\mathsf{P}_{Y|\vartheta}$ disagree
  - Define $H_\vartheta(t) = \mathsf{P}_{Y|\vartheta}(T_{Y,\vartheta} > t)$.
  - Then $H_\theta(T_{Y,\theta}) \sim \text{Unif}(0,1)$ when $Y \sim \mathsf{P}_{Y|\theta}$.
- This can be used to construct procedures in a natural way:
  - Reject $H_0 : \theta \in A$ iff $\sup_{\vartheta \in A} H_\vartheta(T_{y,\vartheta}) \leq \alpha$.
  - $C_\alpha(y) = \{\vartheta : H_\vartheta(T_{y,\vartheta}) > \alpha\}$
- What about "uncertainty quantification"?

Frequentist to IM, cont.

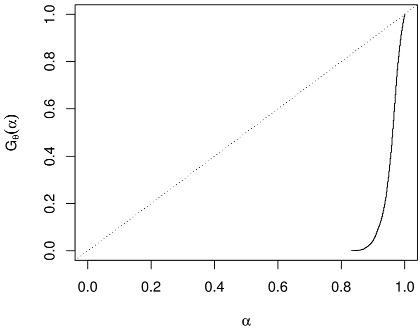
- How to convert these procedures into a valid IM?
- Can the valid IM return probabilities?
- The false confidence theorem says no, but let's check.
- The following steps give a fiducial or confidence distribution:
  - Reinterpret θ as a "random variable."
  - Find Π_y such that $\theta \sim \Pi_y$ implies $H_\theta(T_{y,\theta}) \sim \mathsf{Unif}(0,1)$.
- This implies $\Pi_y\{C_\alpha(y)\} = 1 - \alpha$.
- Validity fails because the probability calculus involved in using Π_y to make belief assignments isn't designed to preserve the sampling distribution properties in $Y \mapsto H_\theta(T_{Y,\theta})$.

Example: Fieller–Creasy

- Bivariate normal: $Y \sim \mathsf{P}_{Y|\theta} = \mathsf{N}_2(\theta, I)$, with $\theta = (\theta_1, \theta_2)$.
- Fiducial/confidence distribution: $\Pi_y = \mathsf{N}_2(y, I)$.
- Goal is to quantify uncertainty about the ratio θ_1/θ_2.
- Let $A = \{\vartheta : \vartheta_1/\vartheta_2 \leq 9\}$.
- Suppose $\theta = (1, 0.1)$, so that $\phi = 10$ and A is false.
- We'd expect $\Pi_Y(A)$ to be small.
- Let's see...

Fieller–Creasy, cont.

- Define the CDF: $G_\theta(\alpha) = \mathsf{P}_{Y|\theta}\{\Pi_Y(A) \leq \alpha\}$.
- Plot uses $\theta = (1, 0.1)$, so $A = \{\phi \leq 9\}$ is false.
- Validity requires the CDF be on or above the diagonal line.
- Note: $\Pi_Y(A)$ is actually close to 1!



- Of course, one's not obligated to use an additive IM.
- In fact, the signs that a non-additive IM is needed are already there in the frequentist theory.
- All we have to do is consider the instinctive manipulations one does to frequentist objects.
- These manipulations are for *marginalization*.
- That is, let θ be the model parameter but suppose interest is in $\phi = \phi(\theta)$, a low-dim feature.
- Both the satellite collision and Fieller–Creasy examples involve marginalization, as do so many others.
- Goal is to quantify uncertainty about ϕ.

Frequentist to IM, cont.

- Let $\pi_y(\vartheta)$ be a p-value for testing a point null about θ.
- A point null about ϕ is a composite null about θ, so the former p-value is

$$\varphi \mapsto \sup_{\vartheta : \phi(\vartheta) = \varphi} \pi_y(\vartheta).$$

- Similarly, a confidence region for ϕ is

$$\Bigl\{ \varphi : \sup_{\vartheta : \phi(\vartheta) = \varphi} \pi_y(\vartheta) > \alpha \Bigr\}.$$

- Notes:
  - the natural operation is *maximizing*
  - it's natural because it preserves the frequentist properties
  - maximizing is a familiar imprecise prob operation.

Frequentist to IM, cont.

- This leads to the following construction of a valid IM.
  - Define a function $\pi_y(\vartheta)$ with the property that

$$Y \sim \mathsf{P}_{Y|\theta} \implies \pi_Y(\theta) \sim \mathsf{Unif}(0,1).$$

  - If $\sup_\vartheta \pi_y(\vartheta) = 1$ for all y, then set

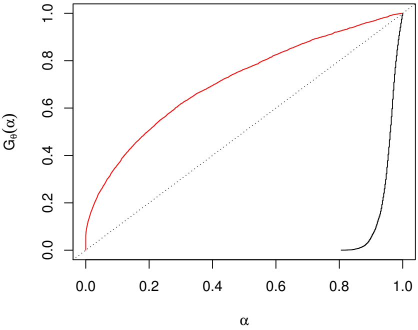
$$\overline{\Pi}_y(A) = \sup_{\vartheta \in A} \pi_y(\vartheta).$$

- This capacity has the mathematical form of a
  - *consonant plausibility function*
  - or, equivalently, a *possibility measure*.
- Validity proof is immediate:

$$\theta \in A \text{ and } \overline{\Pi}_Y(A) \le \alpha \implies \pi_Y(\theta) \le \alpha.$$

Fieller–Creasy, cont.

- Repeat the same simulation experiment as before.
- Compare the CDF of $\Pi_Y(A)$ (black) with that of $\underline{\Pi}_Y(A)$ (red).
- Assertion A is false.
- Again, on or above the diagonal line means valid.



Remarks

- Take-away:
  - "coverage + additivity" doesn't lead to validity,
  - but "coverage + consonance" does.
- There's no cost to choosing imprecision here.
- Importantly, this highlights a fundamental connection between frequentist ideas and imprecise probability.
- Moreover, it's a very special kind of imprecise probability.
- What's special about consonance is that it's "simple" — the IM is fully determined by the contour $\pi_y(\vartheta)$.
- No claims that consonance is the only way to achieve validity, but that's the direction I'm leaning...

Remarks, cont.

- Interpretation of frequentist notions is non-trivial.
- The imprecise prob connection gives a natural interpretation that's consistent with how they're used in practice.
- That is, given the data, posited model, ...:
  - a p-value measures the *plausibility* of the null hypothesis
  - a confidence region is an aggregation of those sufficiently *plausible* parameter values.
- I teach this interpretation to students who know nothing about imprecise probability:

  > *Student: "I wanted to let you know that your discussion of plausibility in ST503 made it easier for me to explain what a p-value is when I was interviewing. Thank you for the new perspective on the subject!"*

A characterization result

- The characterization of frequentist inference via imprecise probability above is relatively weak.
- It takes p-values/confidence regions as the primitive, and constructs a valid IM via "coverage + consonance."
- But the imprecise prob connection is not an afterthought or an embellishment; it's already in those procedures.
- The following result makes this (a little) more precise.

A characterization result, cont.

Theorem.

Let $\{C_\alpha : \alpha \in [0,1]\}$ be a family of confidence regions for $\phi = \phi(\theta)$ that satisfies the following properties:

- Coverage. $\inf_\theta \mathsf{P}_{Y|\theta}\{C_\alpha(Y) \ni \phi(\theta)\} \geq 1 - \alpha$ for all α;
- Nested. if $\alpha \leq \beta$, then $C_\beta \subseteq C_\alpha$;
- Compatible. ....

There exists a valid IM for θ, determined by consonance and contour $\pi_y : \Theta \to [0,1]$, such that the marginal plausibility regions for ϕ,

$$C_\alpha^\star(y) = \left\{\varphi : \sup_{\vartheta:\phi(\vartheta)=\varphi} \pi_y(\vartheta) > \alpha\right\}$$

satisfy

$$C_\alpha^\star(y) \subseteq C_\alpha(y) \quad \text{for all } (y, \alpha) \in \mathbb{Y} \times [0,1].$$

M. (2021). `R1/articles/21.01.00002`

A characterization result, cont.

- There's a parallel result for hypothesis tests; see paper.
- "Compatibility" is related to stuff in Part 2.
- What's missing from the conclusion is that there's more structure in the valid IM, also related to stuff in Part 2.
- Take-away messages:
  - Imprecise prob is fundamental to frequentist inference.
  - Plausibility is the *right* way to interpret p-values and confidence, both intuitively and mathematically.
  - Every "good" procedure corresponds to a valid IM (with the structure discussed in Part 2).
  - Proof is constructive...

Part 1 conclusion

- Recap:
  - IMs are general constructs for probabilistic inference
  - defined the *validity* property
  - what mathematical structures are consistent with validity?
  - probability isn't, consonant plausibility is
- Shows how important imprecise prob is to statistical inference, even if the imprecision isn't part of the model.
- New opportunities for the imprecise prob community?

Part 1 conclusion, cont.

- Part 1 mainly focused on the conversion of a good procedure to a valid IM.
- This approach is interesting and useful, but there's no need to start with a procedure.
- That is, valid IMs can be constructed directly — that's mostly what the book is about.
- Preview of Part 2:
  - direct construction of valid IMs
  - examples
  - beyond validity: efficiency considerations

(Optional) Preachy remark 1

- *Why should we care about false confidence?*
- Something won't necessarily go wrong, but there's a risk.
- Statisticians directly involved in the data analysis might be able to manage their risk reasonably well.
- But this level of involvement is rare — academic statisticians develop methods/software to be used by others.
- To manage risk under this increased exposure, either
  - inform users which hypotheses are dangerous,
  - or avoid false confidence altogether.

(Optional) Preachy remark 2

- *Why care about all assertions?*
- There are reasons not to care about all assertions; see Part 3.
- But short of being able to explicitly say what assertions aren't relevant, we need to be cautious.
- Some statisticians want to dismiss the challenging assertions because they're "impractical."
- From a foundational perspective, it's these challenging assertions that provide insights.
- Analogy: mathematicians can appreciate measurability because they've studied non-measurable things.

(Optional) Preachy remark 3

- *What's the cost of validity?*
- To achieve validity, one must be conservative with respect to some (actually many) assertions.
- I don't think this is a bad thing.
- Analogy: if I want to avoid being wrong, then I shouldn't answer a question about zoology with the same degree of precision I would a question about statistics.
- That is, to avoid being wrong when I lack information, I have to be allowed to say "I don't know."
- But I do care about efficiency, that'll be in Part 2.

The end

End of Part 1

Part 2: Intro

- In Part 1 I tried to convince you that valid probabilistic inference requires imprecise probability.
- That argument was based on an attempt to convert p-values and confidence regions into valid inferential models (IMs).
- But that's not the only way, IMs can be constructed directly.
- In Part 2, I'll describe
  - the general construction of a valid IM
  - and how to fine-tune it for efficiency.

Part 2: Outline

- basic IM construction
- validity theorem
- examples
- beyond validity: efficiency
- dimension reduction strategies:
  - conditioning
  - marginalization
- brief comments: generalized IMs and prediction
- look back at the characterization from Part 1

Problem setup

- Observable data: Y in a sample space $\mathbb{Y}$.
- Data are assumed to be relevant to the phenomenon under investigation, formalized through a *model*.
- Statistical model: a collection of probability distributions

$$\mathscr{P} = \{\mathsf{P}_{Y|\vartheta} : \vartheta \in \Theta\}.$$

- I write ϑ for generic values of the parameter, and θ for the unknown true value.
- Goal is to quantify uncertainty about the unknown θ based on observed value $Y = y$ of the data.

Problem setup, cont.

Definition.

An *inferential model* (IM) is a mapping from $(y, \mathscr{P}, \ldots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^\Theta \to [0,1]$ such that, for each $A \subseteq \Theta$, $\underline{\Pi}_y(A)$ represents ones degree of belief in A, given data, etc.

- Quantify uncertainty via an inferential model (IM)
- We want that IM to be reliable or *valid*.
- Part 1: validity requires non-additivity, imprecision.
- Question: *how to construct such a thing directly?*

IM construction

- Express the statement "$Y \sim \mathsf{P}_{Y|\theta}$," as

$$Y = a(\theta, U), \quad U \sim \mathsf{P}_U, \quad (a, \mathsf{P}_U) \text{ known.}$$

  (Notation: auxiliary variable U takes values in $\mathbb{U}$.)

- This describes how to simulate from the model, but also provides some relevant insight for inference.[11]
- Intuition: *If U were observable*, then just solve for θ in terms of the observed Y and U values — problem solved!
- Unfortunately, *U is not observable*...
- The distribution of U is known, so we can "guess" its unobserved value with some degree of reliability.

[11] Chap. 4 in the book; also, `arXiv:1206.4091`

Construction: A-step

- The relation "$Y = a(\theta, U)$" is called an *association*.
- Define the set-valued mapping

$$(y, u) \mapsto \Theta_y(u) = \{\vartheta \in \Theta : y = a(\vartheta, u)\}.$$

- Intuition:
  - if $Y = y$ were observed and an oracle told me that $U = u$,
  - then I would *know* "$\theta \in \Theta_y(u)$,"
  - and that's the strongest possible conclusion.

Construction: P-step

- Goal is to (probabilistically) guess or *predict* the unobserved value of U.
- Critical points:
  - Uncertainty about a new draw from P_U would best be quantified by the *a priori* distribution P_U.
  - The unobserved value of U has a different status, it's fixed but unknown, so P_U isn't right *a posteriori*.
  - This status change implies that imprecise probability is needed to (reliably) quantify uncertainty.
- The P-step proceeds to guess/predict the unobserved value of U with a *random set*,[12] $\mathcal{S} \sim \mathsf{P}_{\mathcal{S}}$, taking values in $2^{\mathbb{U}}$.

[12] This can also be described via possibility measures, see the end of Part 2.

Construction: C-step

- *Combine* the output of the A- and P-steps, with the observed data $Y = y$, to get a new random set

$$\Theta_y(\mathcal{S}) = \bigcup_{u \in \mathcal{S}} \Theta_y(u).$$

- The *IM output* is the distribution of $\Theta_y(\mathcal{S})$ wrt $\mathcal{S} \sim \mathsf{P}_{\mathcal{S}}$.

IM output

- Key summaries:[13]
  - Belief/lower probability
$$\underline{\Pi}_y(A) = \mathsf{P}_{\mathcal{S}}\{\Theta_y(\mathcal{S}) \subseteq A\}, \quad A \subseteq \Theta.$$
  - Plausibility/upper probability
$$\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c) = \mathsf{P}_{\mathcal{S}}\{\Theta_y(\mathcal{S}) \cap A \neq \varnothing\}, \quad A \subseteq \Theta.$$
- The plausibility contour is also important:
$$\pi_y(\vartheta) = \mathsf{P}_{\mathcal{S}}\{\Theta_y(\mathcal{S}) \ni \vartheta\}, \quad \vartheta \in \Theta.$$

[13] Here I'm assuming that $\Theta_y(\mathcal{S}) \neq \varnothing$ with $\mathsf{P}_{\mathcal{S}}$-probability 1, but this is not necessary; see the end of Part 2.

Remarks

- The user-specified $\mathcal{S}$ is unique to this construction.
- There are similarities with Fisher's fiducial argument and Dempster's extension from the 1960s.
- In particular, if $\mathcal{S}$ is a random singleton, i.e.,

$$\mathcal{S} = \{U\}, \quad U \sim \mathsf{P}_U,$$

  then the IM output agrees with that of Dempster's approach.
- But validity is not guaranteed with singleton $\mathcal{S}$ (see below), so non-singleton sets are needed.
- Effect of choosing non-singleton sets:
  - non-zero gap between lower and upper probabilities
  - gap implies non-additivity, critical to validity

Validity theorem

- Recall, validity ensures that, e.g., beliefs assigned to false assertions do not tend to be large.
- Relevant to reliability of belief assignments.

Definition.

An IM with output $y \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$ is *valid* if

$$\sup_{\theta \notin A} \mathsf{P}_{Y|\theta}\{\underline{\Pi}_Y(A) > 1 - \alpha\} \leq \alpha, \quad \text{all } A \subseteq \Theta, \text{ all } \alpha \in [0,1]$$

or, equivalently, if

$$\sup_{\theta \in A} \mathsf{P}_{Y|\theta}\{\overline{\Pi}_Y(A) \leq \alpha\} \leq \alpha, \quad \text{all } A \subseteq \Theta, \text{ all } \alpha \in [0,1].$$

Validity theorem, cont.

Theorem.

With a "suitable choice of $\mathcal{S} \sim \mathsf{P}_{\mathcal{S}}$," the IM above is valid.

- The IM can be used to design testing and confidence procedures just like discussed above.
- Practical consequences of validity:
  - "Reject $H_0 : \theta \in A$ if $\overline{\Pi}_y(A) \leq \alpha$" is a size α test.
  - The $100(1-\alpha)\%$ plausibility region

$$\{\vartheta : \pi_y(\vartheta) \geq \alpha\}$$

    is a $100(1-\alpha)\%$ confidence region.

"Suitable choice of $\mathcal{S}$"

- For some intuition, what do we want $\mathcal{S}$ to do?
- Let $u_{y,\theta}$ denote the unobserved value of U corresponding to the observed $Y = y$ and true θ.
- Recall: $\Theta_y(\mathcal{S}) = \bigcup_{u \in \mathcal{S}} \Theta_y(u)$.
- Key insight: $\Theta_y(\mathcal{S}) \ni \theta \iff \mathcal{S} \ni u_{y,\theta}$.
- *A priori*, however, $u_{y,\theta}$ is just a realization of $U \sim \mathsf{P}_U$.
- Therefore:
  - if I choose $\mathcal{S}$ to be "good" at predicting realizations from P_U,
  - then, e.g., $\pi_y(\theta)$ at true θ is unlikely to be too small.

"Suitable choice of $\mathcal{S}$," cont.

Definition.

Given $\mathcal{S} \sim \mathsf{P}_{\mathcal{S}}$, define $f(u) = \mathsf{P}_{\mathcal{S}}(\mathcal{S} \ni u)$. Then $\mathcal{S}$ is *valid* if

$$U \sim \mathsf{P}_U \implies f(U) \text{ is stochastically no smaller than Unif}(0,1).$$

- The intuition from the previous slide(s) implies there must be some connection between $\mathsf{P}_{\mathcal{S}}$ and P_U.
- Validity makes this connection mathematically precise.
- Two extreme cases:
  - $\mathcal{S} \equiv \mathbb{U}$, valid.
  - $\mathcal{S} = \{U\}$ with $U \sim \mathsf{P}_U$, not valid.
- The above condition is actually easy to arrange...

"Suitable choice of $\mathcal{S}$," cont.

Lemma.

Let $h : \mathbb{U} \to \mathbb{R}$ be a function such that the distribution of $h(U)$ is non-atomic. Then the random set

$$\mathcal{S} = \{u \in \mathbb{U} : h(u) \leq h(U)\}, \quad U \sim \mathsf{P}_U,$$

is valid in the sense above.

- Common example is $\mathsf{P}_U = \mathsf{Unif}(0, 1)$:
  - good "default" choice
  - $h(u) = |u - 0.5|$.
- Later we'll see that there are reasons to take h to be bowl-shaped, leading to nested random sets.
- This is not the only way to get a valid $\mathcal{S}$...

Examples: binomial

- Let $Y \sim \mathsf{P}_{Y|\theta} = \mathsf{Bin}(n, \theta)$, distribution function F_θ.
- Construct a valid IM for θ:
  A. $F_\theta(Y-1) \leq U < F_\theta(Y)$, $U \sim \mathsf{Unif}(0,1)$.
  P. Simple (but not best) "default" random set

$$\mathcal{S} = \big\{u : |u - 0.5| \leq |U - 0.5|\big\}, \quad U \sim \mathsf{Unif}(0,1).$$

  C. Combine to get[14]

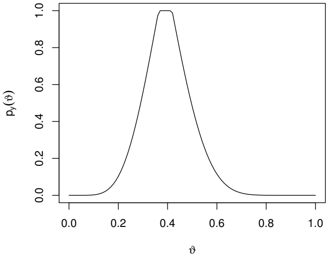
$$\begin{aligned}\Theta_y(\mathcal{S}) &= \bigcup_{u \in \mathcal{S}} \{\theta : F_\theta(y-1) \leq u < F_\theta(y)\} \\ &= \big[1 - G^{-1}_{n-y+1,y}(\tfrac{1}{2} + |U - \tfrac{1}{2}|), 1 - G^{-1}_{n-y,y+1}(\tfrac{1}{2} - |U - \tfrac{1}{2}|)\big],\end{aligned}$$

- Distribution of $\Theta_y(\mathcal{S})$ only depends on $U \sim \mathsf{Unif}(0,1)$...

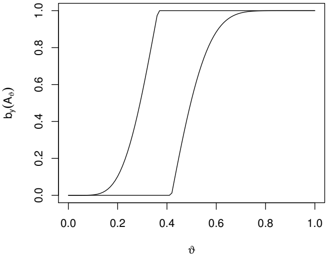
[14] Use fact that $F_\theta(y) = G_{n-y,y+1}(1-\theta)$, beta distribution.

Examples: binomial, cont.

- Data: $n = 18$, $y = 7$.
- Plots of plausibility contour $\pi_y(\vartheta)$ and of the lower and upper probabilities of $A_\vartheta = [0, \vartheta]$.



(e) Plausibility contour



(f) lower/upper probabilities

Examples: general scalar problem

- Problems first considered by Fisher:
  - Scalar $Y \sim \mathsf{P}_{Y|\theta}$ and scalar θ
  - continuous distribution function F_θ
  - range of $F_\theta(y)$ unconstrained by fixed y or fixed θ.
- IM construction:

  A. $Y = F_\theta^{-1}(U)$, $U \sim \mathsf{P}_U = \mathsf{Unif}(0,1)$

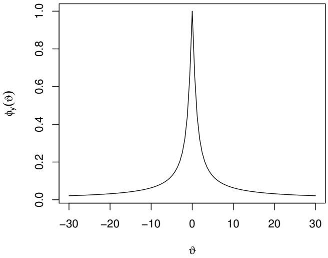
  P. "Default" $\mathcal{S} = \{u \in [0,1] : |u - 0.5| \leq |U - 0.5|\}$, $U \sim \mathsf{Unif}(0,1)$

  C. $\Theta_y(\mathcal{S}) = \{\vartheta : F_\vartheta(y) \in \mathcal{S}\}$, with

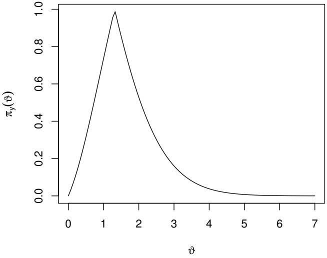
$$\pi_y(\vartheta) = \mathsf{P}_{\mathcal{S}}\{\Theta_y(\mathcal{S}) \ni \vartheta\} = 1 - |2F_\vartheta(y) - 1|.$$

- Lots of examples can be covered by this analysis.

- Two cases:
  - $\mathsf{P}_{Y|\theta} = \mathsf{Cauchy}(\theta, 1)$
  - $\mathsf{P}_{Y|\theta} = \mathsf{Gamma}(\theta, 1)$.
- Plots below show the plausibility contour, $\pi_y(\vartheta)$.



(g) $y = 0$, Cauchy



(h) $y = 1$, gamma

Examples: multi-parameter

- n iid observations from $\mathsf{P}_{Y|\theta} = \mathsf{N}(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$.
- Start with sufficient statistics $(\hat\mu, \hat\sigma^2)$; justification later.
- IM construction:
  A. Let $(U_1, U_2) \overset{\text{iid}}{\sim} \mathsf{Unif}(0,1)$, $G = (n-1)^{-1}\mathsf{ChiSq}(n-1)$, and $\Phi = \mathsf{N}(0,1)$. Then

  $$\hat\mu = \mu + \sigma n^{-1/2}\Phi^{-1}(U_1) \quad \text{and} \quad \hat\sigma^2 = \sigma^2 G^{-1}(U_2).$$

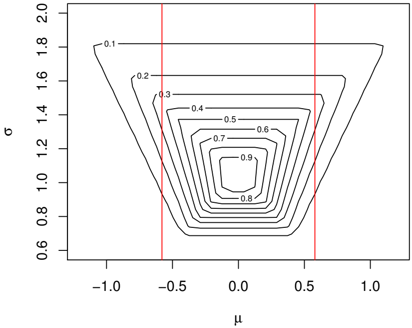
  P. Nested, 2-dim analogue of the "default"

  $$\mathcal{S} = \Big\{(u_1, u_2) : \max_{i=1,2} |u_i - 0.5| \leq \max_{i=1,2} |U_i - 0.5|\Big\}.$$

  C. Sort of messy...

Examples: multi-parameter, cont.

- Data: $n = 10$ and $(\hat{\mu}, \hat{\sigma}^2) = (0, 1)$.
- Plot shows:
  - joint plausibility contour for (μ, σ^2)
  - red lines: (marginal) 90% Student-t interval for μ
- This is valid but inefficient — to match the best inference on μ, we need to do formal marginalization...



Beyond validity: efficiency

- I've focused mainly on validity so far, but this isn't the only practical consideration.
- Recall:
  - Validity can be achieved simply by taking $\mathcal{S} \equiv \mathbb{U}$.
  - Results in $\overline{\Pi}_y(A) \equiv 1$ for all $A \neq \varnothing$.
- Obviously, an IM with vacuous output is useless.
- So it amounts to a balancing act:
  - validity requires $\mathcal{S}$ to be sufficiently large (in some sense)
  - but being too large causes in inefficiency.
- The goal is to take $\mathcal{S}$ as small as possible (in some sense) while maintaining validity.

Efficiency, cont.

- Some progress on efficiency and optimality has been made, but the theory is still incomplete.
- One important result:[15]
  - roughly, for any valid random set, there exists another that's valid, *nested*, and no less efficient;
  - therefore, I only use nested random sets.
- I don't want to go through these efficiency and optimality results formally here now.
- There's a more important and related consideration concerning the *dimension* of the auxiliary variable.

[15] Theorem 4.3 in the book

Efficiency, cont.

- *Curse of dimensionality:* roughly, everything is more difficult in high dimensions than it is in low dimensions.
- Suppose data $Y = (Y_1, \ldots, Y_n)$ consists of n iid observations, with distribution depending on a low-dim θ.
- The naive association "$Y = a(\theta, U)$" involves a n-dimensional auxiliary variable $U = (U_1, \ldots, U_n)$
- Often n will be large, which presents the challenge of guessing the high-dim U with a random set.
- However:
  - it is generally possible to reduce the U dimension
  - simplifies the job of choosing valid/efficient random sets.

Dimension reduction: conditioning

- As a simple illustration, suppose we have n data points from a location model, i.e., $Y_i = \theta + U_i$, $i = 1, \ldots, n$.
- After some algebraic manipulations, the above system of equations is equivalent to

$$\begin{aligned} \bar{Y} &= \theta + \bar{U} \\ Y_i - \bar{Y} &= U_i - \bar{U}, \quad i = 1, \ldots, n. \end{aligned}$$

- Key observation: while U is unobservable, *there is a feature of U, namely $\{U_i - \bar{U} : i = 1, \ldots, n\}$, that is observable.*
- Consequences:
  - make a change of variable $U \mapsto \bar{U}$ to reduce dimension
  - use info about observed features to sharpen prediction of $\bar{U}$

- Abstract this idea...[16]
- Suppose there exists bijections $y \mapsto (T(y), H(y))$ and $u \mapsto (\tau(u), \eta(u))$ such that "$Y = a(\theta, U)$" is equiv to

$$T(Y) = b(\theta, \tau(U)) \quad \text{and} \quad H(Y) = \eta(U).$$

- Want $T(y)$ and $\tau(u)$ to have same dimension as θ.
- Key point: the second expression has no θ in it, hence the value of $\eta(U)$ is *observable*.
  - We don't need to guess $\eta(U)$, hence dimension reduction
  - Observed $\eta(U)$ helps improve guess of unobserved $\tau(U)$
- Conditional association:

$$T = b(\theta, V), \quad V \sim \mathsf{P}_{\tau(U) \mid \eta(U) = H(y)}.$$

[16] Chap. 6 in book; also `arXiv:1211.1530`

Conditioning, cont.

- How to find (T, H) and (τ, η)?
- Sufficient statistics:
  - For regular problems, min suff stat T has same dim as θ
  - Just use an association for T, no conditioning needed.
- Group invariance:
  - T is an equivariant estimator and $\tau = T$
  - H is a maximal invariant and $\eta = H$.
- Solving a partial differential equation...

Conditioning, cont.

- Let $u_{y,\theta}$ be a solution to the equation $y = a(\theta, u)$.
- $\eta(U)$ would be observed if $\eta(u_{y,\theta})$ is constant in θ.
- That is, we seek η that satisfies

$$\frac{\partial \eta(u_{y,\theta})}{\partial \theta} = 0.$$

- I don't know about existence in general, but sometimes I can solve it, e.g., method of characteristics.
- Useful in non-regular problems when sufficiency doesn't help.

Conditioning, cont.

- Same location model illustration: $Y = \theta 1_n + U$.
- $u_{y,\theta} = y - \theta 1_n$.
- Goal is to solve

$$0 = \frac{\partial \eta(u_{y,\theta})}{\partial \theta} = \frac{\partial \eta(u)}{\partial u}\Big|_{u=u_{y,\theta}} \cdot \frac{\partial u_{y,\theta}}{\partial \theta}.$$

- Right-most factor is 1_n in this case.
- Need η such that $\partial \eta(u)/\partial u$ sends constant vectors to 0.
- Take $\eta(u) = Mu$ where, e.g., $M = I_n - n^{-1} 1_n 1_n^\top$.
- Then $U_i - \bar{U} = Y_i - \bar{Y}$ are *observed*.

Conditioning, cont.

- All the "standard" examples can be covered by one or more of the techniques above.
- What about "non-standard" examples, is there anything new that this perspective offers?
- Consider a so-called *non-regular* problem, where the dimension reduction via sufficiency is unsatisfactory.
- e.g., $\mathsf{N}(\theta, \theta^2)$, $\mathsf{Unif}(\theta, \theta + 1)$, etc.
- An extension of the conditioning approach above can reduce dimension further, for valid and efficient inference.
- Based on a *localization* technique.

- Fix a localization point $t \in \Theta$.
- Re-express the baseline association as

$$T(Y) = b(\theta, \tau(U)) \quad \text{and} \quad H_t(Y) = \eta_t(U).$$

- As above, we can construct a t-dependent IM which is valid at $t = \theta$. Specifically:
  - Let $\pi_y(\cdot \mid t)$ denote the t-dependent plausibility contour.
  - Then $\mathsf{P}_{Y|\theta}\{\pi_Y(\theta \mid \theta) \le \alpha\} \le \alpha$, for all $\theta \in \Theta$.
- Assuming that $\sup_\vartheta \pi_y(\vartheta \mid \vartheta) = 1$ for each y, we can use consonance to construct a *fused* plausibility function[17]

$$\overline{\Pi}_y^{(\mathrm{F})}(A) = \sup_{\vartheta \in A} \pi_y(\vartheta \mid \vartheta).$$

[17] This is an updated explanation compared to that in the paper/book.

Conditioning, cont.

- A variation on the above location model:[18]

$$Y_1 = \theta + U_i, \quad Y_2 = \theta^{1/3} + U_2, \quad U_1, U_2 \overset{\text{iid}}{\sim} \mathrm{N}(0, 1).$$

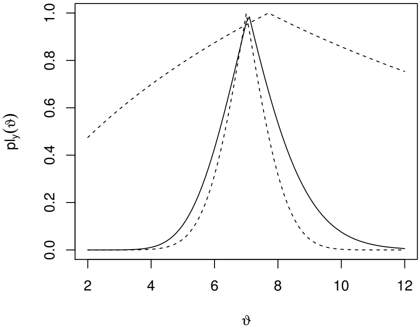
- "Non-regular," no reduction via sufficiency is possible.
- Same PDE as before can be set up, but solving it seems to require the localization trick.
- Skipping details, the localized plausibility contour is

$$\pi_y(\vartheta \mid \vartheta) = 1 - \Bigl| 2\Phi\Bigl(\frac{y_1 - \vartheta + \frac{1}{3\vartheta^{2/3}}(y_2 - \vartheta^{1/3})}{(1 + \frac{1}{9\vartheta^{4/3}})^{1/2}} \Bigr) - 1 \Bigr|.$$

[18] e.g., Seidenfeld (1992), *Statist. Sci.*

Conditioning, cont.

- Cube-root model, true value is $\theta = 7$.
- Plot plausibility contours based on individual observations (dashed) and the one based on fusion (solid).



Conditioning, cont.

- Classical example: bivariate standard normal, correlation θ.
- Non-regular: parameter is 1-dim but min suff stat is 2-dim.
- Table below compares coverage probability and lengths of three 90% confidence intervals for θ.

	Coverage probability			Expected length		
n	IM	$r^\star$	Bayes	IM	$r^\star$	Bayes
10	0.896	0.845	0.880	0.66	0.61	0.62
25	0.895	0.867	0.883	0.42	0.40	0.41
50	0.907	0.897	0.907	0.30	0.30	0.30
100	0.903	0.888	0.896	0.21	0.21	0.21

Conditioning, cont.

- Dimension reduction via conditioning is an effective strategy for constructing a valid and efficient IM.
- Works for standard examples, and provides new insights on how to handle non-standard examples.
- References using these techniques:
  - Cheng, Gao, and M. (2014), `arXiv:1406.3521`
  - M. and Lin (2016), `arXiv:1608.06791`
  - M. and Syring (2019), `R1/article/2019-03-5`

Dimension reduction: marginalization

- Consider a decomposition $\theta = (\phi, \lambda)$ where
  - ϕ is the *interest parameter*
  - λ is the *nuisance parameter*
- This is a common situation, e.g., we often care about a mean more so than we do the variance.
- If we only care about inference on ϕ:
  - creates an opportunity for dimension reduction
  - improved efficiency
- This is called *marginalization*, and the goal is to construct a marginal IM for ϕ.

Marginalization, cont.

- Start with an association $Y = a(\theta, U)$ for θ, where U has the same dimension as θ, maybe post-conditioning.
- Suppose it can be rewritten as

$$R(Y, \phi) = c(\phi, V_1) \quad \text{and} \quad d(Y, V_2, \phi, \lambda) = 0.$$

- Suppose $d(\cdot)$ satisfies

$$\forall\ (y, v_2, \phi) \quad \exists\ \lambda \quad \text{such that } d(y, v_2, \phi, \lambda) = 0.$$

- Marginal association for ϕ: $R(Y, \phi) = c(\phi, V_1)$.
- Dimension reduction by *ignoring the d component*.[19]

[19] Chap. 7 in the book; also, `arXiv:1306.3092`

Marginalization, cont.

- Let $Y = (Y_1, \ldots, Y_n)$ be iid $\mathrm{N}(\mu, \sigma^2)$, target $\phi = \mu$.
- Conditioning $\to$ suff stats $(\hat{\mu}, \hat{\sigma}^2)$.
- Normal mean problem, revisited:
  - Reduced association

$$\hat{\mu} = \mu + \sigma n^{-1/2} U_1 \quad \text{and} \quad \hat{\sigma} = \sigma U_2,$$

    where $U_1 \sim \mathrm{N}(0,1)$, $U_2 \sim \{(n-1)^{-1}\mathrm{ChiSq}(\mathrm{n}-1)\}^{1/2}$.
  - Rewrite this as

$$n^{1/2}(\hat{\mu} - \mu)/\hat{\sigma} = U_1/U_2 \quad \text{and} \quad \hat{\sigma} = \sigma U_2.$$

  - Ignore the second equation to get the marginal assoc

$$\hat{\mu} = \mu + \hat{\sigma} n^{-1/2} V, \quad V \sim \mathrm{t}_{n-1}.$$

- Efficient!

Marginalization, cont.

- Fieller–Creasy: $Y_i \overset{\text{ind}}{\sim} \mathsf{N}(\theta_i, 1)$, $i = 1, 2$.
- Target is $\phi = \theta_1/\theta_2$; $\lambda = \theta_2$ is nuisance.
- Following Chapter 7.3.2 in the book:
  - Baseline association

$$Y_1 = \phi\lambda + U_1 \quad \text{and} \quad Y_2 = \lambda + U_2.$$

  - Rewrite it as

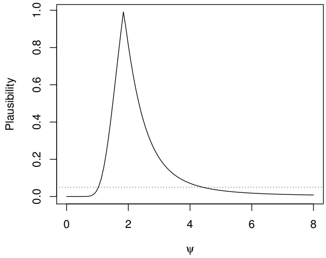
$$\frac{Y_1 - \phi Y_2}{(1+\phi^2)^{1/2}} = \underbrace{\frac{U_1 - \phi U_2}{(1+\phi^2)^{1/2}}}_{=V\sim\mathsf{N}(0,1)} \quad \text{and} \quad \underbrace{Y_2 - \lambda - U_2 = 0}_{\text{ignore!}}.$$

  - Simple reduced association

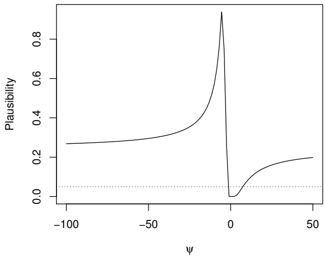
$$\frac{Y_1 - \phi Y_2}{(1+\phi^2)^{1/2}} = V, \quad V \sim \mathsf{P}_V = \mathsf{N}(0,1).$$

Marginalization, cont.

- Plots show plausibility contours for two data sets
- Second plot has a very unusual shape:
  - plausibility interval is unbounded
  - this feature is necessary for validity[20]
- This IM solution matches the best possible.



(i) $y = (6.38, 3.46)$



(j) $y = (5.98, -1.16)$

[20] Gleser & Hwang (1987), *Annals of Statistics*

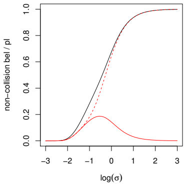
Marginalization, cont.

- Many-normal-means: $Y \sim \theta + U$, where $U \sim \mathsf{N}_n(0, I)$.
- $\theta = \phi\lambda$, where $\phi = \|\theta\|$ and $\lambda = \phi^{-1}\theta \in \{\text{unit } n\text{-sphere}\}$.
- $n = 2$ case $\iff$ simple satellite collision problem
- From Chapter 7.3.3 in the book, the reduced association is

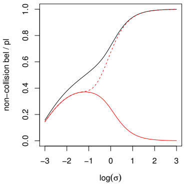
$$\|Y\|^2 = F_{n,\phi}^{-1}(V), \quad V \sim \mathsf{P}_V = \mathsf{Unif}(0,1),$$

  where $F_{n,\phi} = \mathsf{ChiSq}(n; \phi^2)$.
- Looks simple, but there are some constraints that need to be handled carefully; I won't discuss this here.

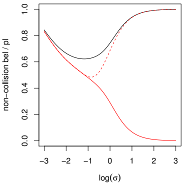
- Same satellite collision illustration as before, now with
  - Π_y(non-collision) (black)
  - $\underline{\Pi}_y$(non-collision) (red) and $\overline{\Pi}_y$(non-collision) (red, dashed).
- Gap between belief and plausibility is increasing in σ.



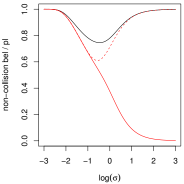
(k) $\|y\|^2 = 0.5$



(l) $\|y\|^2 = 0.9$

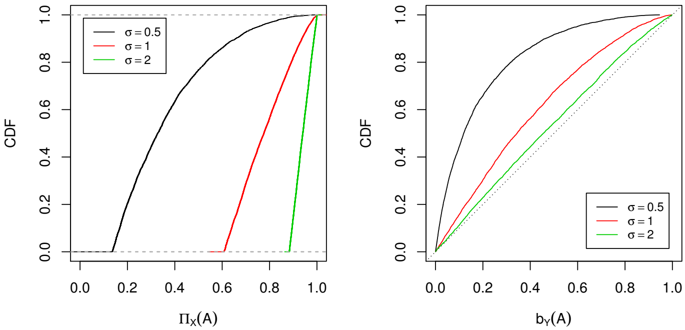


(m) $\|y\|^2 = 1.1$



(n) $\|y\|^2 = 1.5$

- Check validity by evaluating CDF of prob/belief.
- Same experiment as before.
  - Left: Π_Y(non-collision) tends to be large
  - Right: $\underline{\Pi}_Y$(non-collision) tends to be not large



Marginalization, cont.

- To recap:
  - marginal inference, when only ϕ is of interest
  - dimension reduction improves efficiency
  - lots of non-trivial examples where this marginalization applies
- Sometimes the aforementioned condition on the d-component doesn't hold, e.g., Behrens–Fisher.
- We developed a generalized marginal IM to handle B–F and other examples, but I won't discuss it here.
- I don't believe this generalization has been investigated beyond those examples in the paper.

Generalized IM construction

- Conditioning can be difficult to implement in applications.
- Alternative strategy to dimension reduction is to forget about writing an association that fully captures the sampling model.
- Just write the association in terms of some low-dim summary.
- Let $R_{Y,\theta}$ be a real-valued function of (Y, θ).
- Example: $R_{y,\vartheta} = L_y(\vartheta)/L_y(\hat{\theta}_y)$, likelihood ratio.
- If large values of $R_{y,\vartheta}$ indicate agreement between y and ϑ, then define

$$\pi_y(\vartheta) = \mathsf{P}_{Y|\vartheta}\{R_{Y,\vartheta} > R_{y,\vartheta}\}.$$

- Get $\underline{\Pi}_y$ and $\overline{\Pi}_y$ using consonance.

Generalized IM, cont.

- Conceptually simple, but computation can be non-trivial.
- References:
  - M. (2015), `arXiv:1203.6665`
  - M. (2018), `arXiv:1511.06733`
  - Cahoon and M. (2020), `R1/article/2019-09-25`
  - Cahoon and M. (202x), `R1/article/2019-11-17`
  - Cella and M. (2021), *coming soon*
- My gut feeling is that there must be some efficiency lost due to the extreme dimension reduction, but no proof yet.

Prediction problems

- Prediction is an extreme marginalization problem, i.e., all the model parameters are nuisance.
- Can apply the above dimension reduction strategies to construct a valid IM for prediction.
- New approach draws on the generalized IM idea and a connection to Shafer & Vovk's *conformal prediction*.
- References:
  - M. and Lingham (2016), `arXiv:1403.7589`
  - Cella and M. (2021), `R1/article/2020-01-12`

Characterization theorem, revisited

- From Part 1, recall the result that said, roughly, "frequentist procedures can be characterized by valid IMs."
- It turns out there's more that can be said.
- In particular, that valid IM has the mathematical form of those constructed here in Part 2.
- That is, the valid IM in that theorem corresponds to the specification of a random set targeting an auxiliary variable.
- Interesting point is that we generally need something like the *fusion* step described above for non-reg conditioning.

Characterization theorem, cont.

- Consider a *family* of random sets on an auxiliary variable space $\mathbb{U}$, i.e., $\mathcal{S} \sim \mathsf{P}_{\mathcal{S} \mid t}$, distributions indexed by $t \in \Theta$.
- For each t, do the above IM construction and let

$$\pi_y(\vartheta \mid t) = \mathsf{P}_{\mathcal{S} \mid t}\{\Theta_y(\mathcal{S}) \ni \vartheta\}, \quad \vartheta \in \Theta.$$

- Define a *fused* plausibility:
  - Assume $\sup_\vartheta \pi_y(\vartheta \mid \vartheta) = 1$ for each y.
  - Set $\overline{\Pi}_y^{(\mathrm{F})}(A) = \sup_{\vartheta \in A} \pi_y(\vartheta \mid \vartheta)$.
- Easy to check: if each t-specific IM is valid, then so is that derived from their fusion.

Characterization theorem, cont.

Theorem (mostly copied from before)

Let $\{C_\alpha : \alpha \in [0,1]\}$ be a family of confidence regions for $\phi = \phi(\theta)$ that satisfies the following properties:

- Coverage. $\inf_\theta \mathsf{P}_{Y|\theta}\{C_\alpha(Y) \ni \phi(\theta)\} \geq 1 - \alpha$ for all α;
- Nested. if $\alpha \leq \beta$, then $C_\beta \subseteq C_\alpha$;
- Compatible. there exists $\mathbb{U}$ and a compatible association...

There exists a valid IM for θ, *derived from a valid random set (or family thereof) on $\mathbb{U}$ and the IM construction above*, such that

$$C_\alpha^\star(y) = \Big\{\varphi : \sup_{\vartheta:\phi(\vartheta)=\varphi} \pi_y(\vartheta \mid \vartheta) > \alpha\Big\}$$

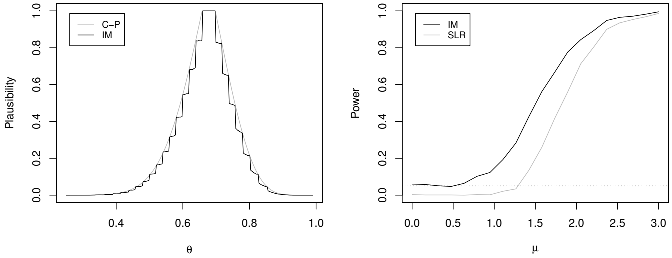
satisfies $C_\alpha^\star(y) \subseteq C_\alpha(y)$ for all $(y,\alpha) \in \mathbb{Y} \times [0,1]$.

M. (2021). `R1/articles/21.01.00002`

Characterization theorem, cont.

- There's a parallel result for hypothesis tests.
- Lots more discussion of *compatibility* can be found in the paper; "simple" sufficient conditions are still missing.
- As before, establishes an important connection between imprecise probabilities and frequentist properties.
- Also, there's apparently something fundamental about the use of suitable random sets on an auxiliary variable space.
- Moreover, the above theorem's proof is *constructive*, which defines an "algorithm" with
  - input: a procedure with frequentist guarantees
  - output: a valid IM with procedures that are no worse

- Left plot shows the Clopper–Pearson plausibility contour along with that of the IM improvement.
- Right plot shows the power of a new split LR test[21] and along with that of the IM improvement.



(o) Binomial plausibility (p) Power, mixture model

[21]Wasserman et al (2020, *PNAS*), `arXiv:1912.11436`

Part 2 conclusion

- Three-step construction of a valid IM:
  A. *associate* the data and unknown parameters with an unobservable auxiliary variable;
  P. *predict* that unobserved value of the auxiliary variable with a valid random set;
  C. *combine* the results of the two previous steps to get an IM (data-dependent lower/upper probability) that is valid.
- For efficiency, it may be necessary to do some dimension reductions before completing the A-step.
- This has proved successful in a number of non-trivial examples, but there are challenges:
  - *conceptual*
  - *computational*

Part 2 conclusion, cont.

- A few other remarks...
- IMs and possibility theory:
  - IMs can be constructed via possibility measures[22]
  - Provides some insight concerning optimality
  - Are the two constructions equivalent?
- What happens if $\Theta_y(\mathcal{S})$ can be empty?
  - One option is to use Dempster's rule of combination to eliminate conflict cases
  - Another option is to manually "stretch" the random set[23]
  - Latter approach was shown to be more efficiency, at least in certain cases involving non-trivial parameter constraints.

[22] Liu and M. (2020), `R1/articles/20.08.00004`

[23] Ermini Leaf and Liu (2012), *IJAR*

The end

End of Part 2

Part 3: Intro

- Parts 1–2 highlighted work that has already been done.
- One of my goals with this series of lectures is to teach you enough that you can start your own projects.
- To help jump-start your efforts, I want to tell you about a few open problems that I think are worth investigating.
- That's the goal of Part 3.

Part 3: Outline

- Open problems/things I'm currently thinking about:
  - Efficiency and optimality
  - Partial prior information
  - False confidence phenomenon
  - Weakening the validity requirement
  - Computations
- Fundamental role of imprecise probability in statistics

Efficiency and optimality

- Assume all the dimension reduction steps have been taken.
- This leaves an unobservable $U \sim \mathsf{P}_U$ whose unobserved value is to be guessed/predicted.
- What's the best random set $\mathcal{S}$ to choose?
- Book gives some optimality results for specific assertions.
- If there's only one A you really care about, then this is fine, but it wouldn't be good for other assertions.
- Questions:
  - If we had A-optimal random sets for every A, would it make sense to "fuse" the individual IMs like before?
  - What about abandoning optimality and considering a more holistic approach, i.e., an "overall good" random set?

Efficiency and optimality, cont.

- A useful result[24] describes how $\mathsf{P}_{\mathcal{S}}$ should allocate mass once its (nested) support $\mathbb{S}$ is given:

$$\mathsf{P}_{\mathcal{S}}(\mathcal{S} \subseteq K) = \sup_{S \in \mathbb{S}: S \subseteq K} \mathsf{P}_U(S).$$

- Then the efficiency/optimality question boils down to one concerning the (nested) support $\mathbb{S} \subset 2^{\mathbb{U}}$.
- My conjecture is that the "best" support is $\mathbb{S}^\star = \{S_\tau\}$,

$$S_\tau = \{u \in \mathbb{U} : p_U(u) \geq \tau\}, \quad p_U = d\mathsf{P}_U / d\lambda.$$

- Connections with possibility theory support this claim.[25]
- In what sense is this choice "best"?

[24] Chap. 4 in book; Thm 4 in M. (2019), `R1/articles/21.01.00002`
[25] Liu and M. (2020), `R1/articles/20.08.00004`

Partial prior information

- Statisticians tend to focus on the two extreme cases:
  - complete prior information
  - no prior information ($\leftarrow$ my focus in Parts 1–2)
- But some problems are in between these two.
- Example: structured high-dimensional problems
  - information about the structure is known (e.g., sparsity)
  - no information about the structure-specific parameters
- Imprecise probability can be used to
  - quantify this partial prior info about θ
  - combine with info from data (e.g., Dempster's rule)
- Combination rules aren't designed to preserve validity![26]
- *How to incorporate prior information, to improve efficiency, without ruining validity?* [27]

[26] M. and Syring (2019), `R1/articles/19.03.00005`
[27] Cella and M. (2019), *ISIPTA proceedings*

False confidence phenomenon

- Recall: *for any IM whose output is additive, there exists false assertions A that tend to be assigned high belief.*
- How serious of a problem is this? In other words, can we characterize the problematic assertions?
- In my experience, the non-trivial examples occur when A is about some non-linear function $\phi = \phi(\theta)$.
    - Satellite collision problem: $\phi = \|\theta\|$
    - Fieller–Creasy: $\phi = \theta_1/\theta_2$
- Other literature describing challenges due to non-linearity:
    - Gleser & Hwang (1987), *Annals of Statistics*
    - Fraser (2011), *Statistical Science*
- Does this help?

Weakening validity

- Validity is a reasonable condition, but it's strong.
- There are a few ways it might be relaxed.
- "All $A \in 2^{\Theta}$" $\to$ "all $A \in \mathcal{A}$, with $\mathcal{A} \subset 2^{\Theta}$"
  - related to false confidence and the previous page
  - pointless to restrict $\mathcal{A}$ severely
  - understanding exactly how much additional flexibility comes by restricting $\mathcal{A}$ will help explain the importance of imprecision
- "All $\alpha \in [0, 1]$" $\to$ "all $\alpha \in [0, \bar{\alpha}]$, with $\bar{\alpha} < 1$"
  - when α can be arbitrarily close to 1, validity requires that $\overline{\Pi}_Y(A)$ can be arbitrarily close to 1 too
  - does this itself imply consonance?
  - Walley (*JSPI* 2002) shows some very interesting results for validity of generalized Bayes with $\bar{\alpha} \ll 1$

Weakening validity, cont.

- New idea based on prediction perspective.[28]
- Suppose Y is observable, goal is to predict $\tilde{Y}$.
- Cella and M. proposed the following definition of validity:

$$\underbrace{\mathsf{E}\big[1\{\overline{\Pi}_Y(A) \leq \alpha\}\, \mathsf{P}(\tilde{Y} \in A \mid Y)\big]}_{\mathsf{P}\{\overline{\Pi}_Y(A) \leq \alpha\,,\, \tilde{Y} \in A\}} \leq \alpha, \quad \text{all } (\alpha, A, \mathsf{P}).$$

- Immediate consequences:
  - $\overline{\Pi}_Y =$ Bayes is valid if P is known
  - $\overline{\Pi}_Y =$ gen Bayes is valid if credal set contains P.
- Not-immediate consequence: *validity implies no sure loss.*
- Extend this idea to the inference problem?

[28] Cella and M. (2020), `R1/articles/20.01.00010`

Computations

- IM developments are intended to be of practical value.
- That is, my goal is to develop ideas and tools that can be used to help solve real-world problems.
- So it's imperative that these things can be computed.
- For all/most of the examples presented here, the computations were relatively easy.
- Same is true for most textbook problems, which covers most what people do in the real-world.
- What about the non-textbook problems?

Computations, cont.

- I know effectively nothing about the extensive work by the imprecise probability community on computation.
- Maybe you already know how to do these things...
- Here is basically what I'm looking for:
  - Simple and efficient Monte Carlo methods for evaluating lower/upper probs based on random sets?
  - Simple and efficient methods for upper prob evaluation via optimization (re: consonance/possibility)?
- Efficiency is important for obvious reasons.
- It's wrong to expect that solutions should be "simple," but practitioners are more likely to use our theory if it's easy to do.

Role of imprecise probability

- The developments presented in this course make clear that imprecise probability is fundamental to statistical inference.
  - not necessarily because models are inherently imprecise
  - but because reliable, data-driven uncertainty quantification requires a certain degree of imprecision.
- What's more, all signs point to *consonant plausibility* as the "right" kind of imprecision for this purpose.
- This is exciting! There's an opportunity to resolve *the most important unresolved problem in statistical inference.*[29]

[29] Quote from B. Efron; see M. (2019), `R1/articles/19.02.00001`, Sec. 5.2

Role of imprecise probability, cont.

- This progress can't be made by the statistics community, largely because no one will work on it.
- At least in the US, foundations work is career suicide:[30]
  - journal editorial boards define "impactful research"
  - there's no funding for foundational research
  - without top-tier publications and funding, there's no tenure
- The imprecise probability community, on the other hand, has the right expertise and an open-minded attitude.
- Thanks for the opportunity to connect with your community and to share these developments.
- If you're interested, *let's get to work!*

[30] It was tough for me, I often considered giving up. Ultimately, I did OK, thanks to the right combination of luck and hardheadedness.

The end

End of Part 3

The end – really

- Links to papers, talks, etc. can be found on my website:

  `www4.stat.ncsu.edu/~rmartin/`

- The videos for these lectures will be made available somewhere, check my website for details.
- If you have questions or just want to talk, please feel free to email me at `rgmarti3@ncsu.edu`.
- Please check out `researchers.one`:
  - new virtual conference feature
  - more new things coming soon

Thanks for your attention!